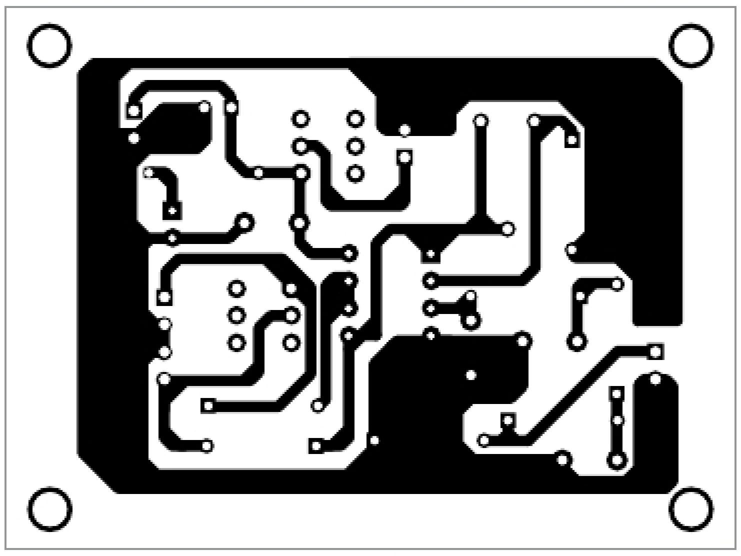


Fig. 3: Actual-size PCB layout of PWM-to-analogue signal converter

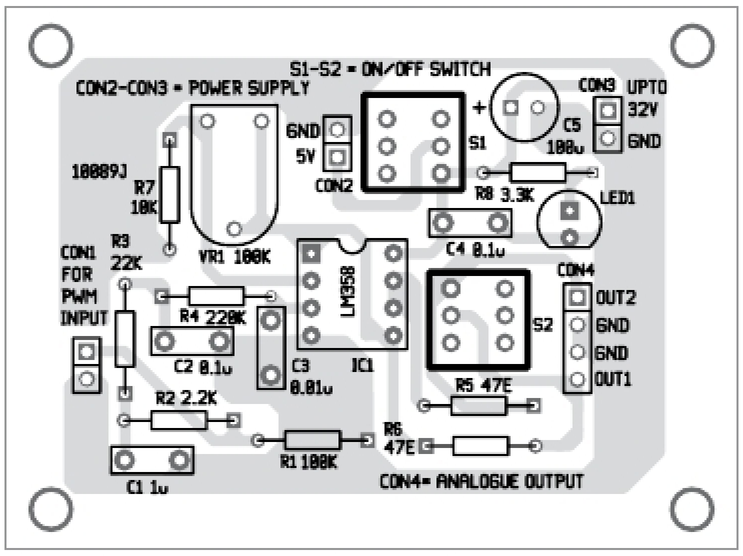


Fig. 4: Components layout for the PCB

EXAMPLES OF REFERENCE VOLTAGES FOR TESTING THE CIRCUIT

Square-wave signal T_{ON}/T, (%)	Approximate Vout as percentage of $Vout_{max}$	Example of Vout, if $Vout_{max}$=5.000V needed	Example of Vout, if $Vout_{max}$=3.000V needed
0	0	0.0V= $Vout_{min}$	0.0V = $Vout_{min}$
10	10	0.5V	0.3V
20	20	1V	0.6V
30	30	1.5V	0.9V
40	40	2.0V	1.2V
50	50	2.5V	1.5V
60	60	3.0V	1.8V
70	70	3.5V	2.1V
80	80	4.0V	2.4V
90	90	4.5V	2.7V
100	100	5.0V = $Vout_{max}$	3.0V = $Vout_{max}$

PARTS LIST

Semiconductors:	
IC1	- LM358, op-amp
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 100-kilo-ohm
R2	- 2.2-kilo-ohm
R3	- 22-kilo-ohm
R4	- 220-kilo-ohm
R5-R6	- 47-ohm
R7	- 10-kilo-ohm
R8	- 3.3-kilo-ohm
VR1	- 100-kilo-ohm preset
Capacitors:	
C1	- 1µF, ceramic disc
C2, C4	- 0.1µF, ceramic disc
C3	- 0.01µF, ceramic disc
C5	- 100µF, 50V electrolytic
Miscellaneous:	
CON1-CON3	- 2-pin connector
CON4	- 4-pin connector
S1-S2	- On/off switch
	- 5V DC power supply

OA2 works as a follower, where gain Av2 = +1

Construction and testing

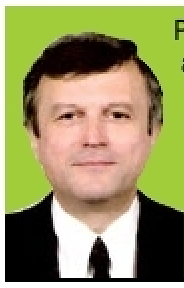
An actual-size PCB layout of the PWM-to-analogue converter is shown in Fig. 3 and its components layout in Fig. 4.

Before testing, note that, maximum output voltages (Vout) at OUT1 and OUT2 depend on: power supply of operational amplifiers, position of potentiometer VR1 and output voltage levels of TTL/CMOS output driving the module.

To test the circuit, fix the frequency of PWM (for example, 1000Hz) and change the duty cycle time period T_{ON} and T_{OFF} of the pulses. Refer to the table to test output voltages.

Linearity of the conversion depends, to some degree, on the parameters of the digital output. A too low or too high PWM duty cycle should be avoided. Usually, duty cycle of PWM <5% and >95% must be avoided. In most cases, PWM DACs are good for driving LEDs, incandescent lamps, LCD displays, DC motors and loudspeakers, among others.

Spectrum of the rectangular signal produced by the PWM is large and variable. This can be a problem without appropriate filtering, construction and PCB layout. Resistors, capacitors and operational amplifiers for filtering and buffering of PWM signals should have appropriate characteristics. Usually with PWM DACs, you obtain high resolutions but not accuracy. EFY



Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFPPT(Casablance), Kingdom of Morocco. Now, he is working as an electronics engineer in the private sector in Bulgaria